Making Ten
Answer Key

Kindergarten–Grade 1   Operations and Algebraic Thinking

Quick Answers

1. 4	4. 3	7. 8	10. 7, 13
2. 7	5. 2, 12	8. 3, 12	
3. 2	6. 4	9. 6, 3	

Curriculum

Level: Kindergarten–Grade 1   Operations and Algebraic Thinking
K.OA.A / 1.OA.C.6 / 2.OA.B.2 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–5 of 5 across 14 tagged checks.

Common wrong answers

3. *If they got 18: added the two numbers instead of taking one away from ten*
5. *If they got 10: filled the frame to ten and forgot the counters still left over*
9. *If they got 4: counted the children who are sitting instead of the seats still empty*

1. Ten-frame with six dots.

Step 1. The top row of a ten-frame is full, so the dots reach 5 at the end of that row, and one more dot in the bottom row makes 6.
Step 2. Count the empty squares instead of starting over: "seven, eight, nine, ten" — that is 4 empty squares.
Step 3. So $6 + \boxed{4} = 10$.
Teaching note: the frame does the work. Once a child sees that a full frame is 10, the empty squares *are* the answer, with no counting on from one.

2. Ten-frame with three dots.

Step 1. Three dots sit in the top row, so the top row still has 2 empty squares.
Step 2. The whole bottom row is empty, which is 5 more, and $2 + 5 = 7$.
Step 3. So $3 + \boxed{7} = 10$.
Listen for: a child who answers 2. That is only the rest of the top row — the bottom row is part of the ten too.

3. Ten cubes, eight taken away.

Step 1. The tower starts as a whole of 10, and taking cubes off asks for the part that is left, so this is a subtraction.
Step 2. Take 8 from 10. Because 8 and 2 are the parts of ten, the leftover is the partner of 8.
Step 3. $10 - 8 = \boxed{2}$ cubes.
Common wrong answer: 18 — the two numbers were added instead of one being taken away from the other.

4. Ten stickers in two pockets.

Step 1. The two pockets hold all 10 stickers between them, so the left pocket and the right pocket are the two parts of ten.
Step 2. One part is 7. Its partner fills the frame: "eight, nine, ten" is 3 more.
Step 3. The right pocket holds $\boxed{3}$ stickers, and $7 + 3 = 10$.

5. Eight on the frame, four in her hand.

(a) Step 1. Filling the frame first is what makes this easy, so start by asking how many empty squares the frame has.
Step 2. Eight dots leave 2 empty squares, so Ellie moves $\boxed{2}$ counters from her hand into the frame.
(b) Step 3. The frame is now full, which is 10, and $4 - 2 = 2$ counters are still in her hand.
Step 4. Count on from ten: "eleven, twelve". So $8 + 4 = 10 + 2 = \boxed{12}$.
Common wrong answer: 10 — the frame was filled and the leftover counters were forgotten. Ask: "is your hand empty?"

6. Ten eggs, six used.

Step 1. The carton is a ten, so the eggs used and the eggs left are the two parts of ten again — the same picture as the cubes, in a new story.
Step 2. 6 and 4 are partners to ten.
Step 3. $10 - 6 = \boxed{4}$ eggs are left.

7. Number bond: whole 10, one part 2.

Step 1. A number bond shows a whole and its two parts, so the missing part is whatever joins 2 to make 10.
Step 2. Count on from 2 to 10: "three, four, five, six, seven, eight, nine, ten" — that is 8 steps.
Step 3. The other part is $\boxed{8}$, and $2 + 8 = 10$.
Teaching note: problems 1, 4 and 7 are the same idea in three pictures — a frame, a story and a bond. Say so out loud; that is the connection the sheet is building.

8. $7 + 5$ by making a ten.

(a) Step 1. Seven dots fill the top row and two of the bottom row, so the frame has 3 empty squares — that is how much of the 5 is needed to reach ten.
Step 2. Move $\boxed{3}$ of the five counters into the frame.
(b) Step 3. The frame is full at 10, and $5 - 3 = 2$ counters are left over.
Step 4. Count on: $10 + 2$, so $7 + 5 = \boxed{12}$.
Why this method: adding to a ten is easy, and breaking the 5 into $3 + 2$ turns a hard fact into an easy one. This is the same move that later becomes carrying.

9. Ten seats at story time.

(a) Step 1. Every seat is either full or empty, so the children and the empty seats are the two parts of ten.
Step 2. $10 - 4 = \boxed{6}$ empty seats.
(b) Step 3. Three more children take three of those empty seats. Either take 3 from the 6 that were empty, or take all 7 children from 10 at once — both give the same count.
Step 4. $10 - 4 - 3 = \boxed{3}$ empty seats.
Common wrong answer: 4 — the children sitting were counted instead of the seats still empty. Point at the chairs.

10. Put it together: Sam's stickers.

(a) Step 1. Sam starts with a whole of 10 and gives some away, so the first question is a take-from-ten.
Step 2. $10 - 3 = \boxed{7}$ stickers.
(b) Step 3. Now 7 stickers meet 6 new ones. Make a ten: 7 needs 3 more to reach ten, so break the 6 into $3 + 3$.
Step 4. $7 + 3 = 10$, and 3 are still left over, so count on: "eleven, twelve, thirteen".
Step 5. Sam ends with $\boxed{13}$ stickers.
Grading note: full credit for any correct method, but ask the child to show the ten. The point of the sheet is that ten is the place to stop and regroup, and part (b) is where that habit pays off.